Musfikur Rahaman

PhD Researcher
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PROFESSIONAL SUMMARY

PhD researcher in Information Science at the University of Arkansas at Little Rock. Dissertation research studies temporal change in knowledge-grounded AI — how systems can determine whether the information they rely on remains appropriate to use now — through a topological study of Mamba and Transformers. PhD researcher in Information Science with expertise in Explainable AI (XAI), multimodal AI systems, computer vision, and transformer-based deep learning. Experienced in developing and deploying AI-powered applications using PyTorch, Hugging Face Transformers, FastAPI, and large language models for healthcare AI, fake news detection, and explainable AI solutions. Research interests include trustworthy AI, medical vision-language models, hallucination detection, and multimodal learning. Passionate about bridging research and industry through scalable AI systems and production-ready machine learning applications. Skilled in designing end-to-end machine learning pipelines, integrating data preprocessing, model training, explainability techniques, and cloud-based deployment to deliver scalable and production-ready AI solutions. Published researcher with experience in literature reviews, experimental design, model development, performance evaluation, and scientific writing, contributing to peer-reviewed research in artificial intelligence, machine learning, bioinformatics, and data analytics.

PyTorch | TensorFlow | Keras | HuggingFace Transformers | Scikit-learn | NumPy | Pandas | Matplotlib | Python | SQL | C++ | Java | FastAPI | REST APIs | Git | GitHub | GitLab | Jupyter Notebook | VS Code | MySQL | MongoDB | PostgreSQL | SQL Server | Supabase | Render | ViT | TimeSformer | BERT | DistilBERT | LLaMA

EDUCATION

University of Arkansas at Little Rock, Ph.D. in Information Science, Information Science — Present

Arkansas Tech University, M.S. in Information Technology, Information Technology, 4.00/4.00 — Apr 2024

University of Information Technology & Sciences, B.S. in Information Technology, Information Technology, 3.83/4.00 — Apr 2021

EXPERIENCE

Graduate Assistant — Little Rock, AR

School of Counseling, Human Performance & Rehabilitation, University of Arkansas at Little Rock — Aug 2024 - Present

Supported departmental operations and faculty research initiatives by developing web-based workflow solutions, managing FERPA-compliant records, leading document digitization efforts, and assisting with public health and minority student research projects within the School of Counseling, Human Performance & Rehabilitation.

- Developed and deployed web-based request and order forms, automating departmental workflows and reducing manual processing overhead
- Led document digitization initiative, converting physical records into structured digital archives to improve data accessibility and compliance
- Supported faculty research on public health disparities, conducting literature reviews and assisting with study design for minority student health studies
- Maintained FERPA-compliant records management across administrative and academic operations

Graduate Assistant — Russellville, AR

Division of Computing Sciences, Arkansas Tech University — Aug 2022 - May 2024

Supported undergraduate computing courses through lab instruction, teaching assistance, and student mentoring in Data Communication & Networking, Software Development, and Data Structures. Assisted students with technical concepts, programming assignments, and academic improvement while actively contributing to research presentations and award-winning graduate research projects.

- Designed and delivered Data Communication & Networking lab sessions; received positive feedback for instructional clarity and engagement
- Served as Teaching Assistant for Software Development and Data Structures courses; provided academic support to 30+ students per semester
- Provided one-on-one mentoring with tailored improvement plans to enhance student understanding and performance
- Presented research at Arkansas Academy of Science, ATU Research Symposium, and LSUS Regional Student Scholars Forum
- Earned 1st Place – Graduate Poster Presentation (Arkansas Academy of Science) and 2nd Place – Graduate Oral Presentation (ATU Research Symposium)

Research Assistant Dhaka, Bangladesh

Department of Information Technology, University of Information Technology & Sciences, Bangladesh Jan 2020 - Apr 2021

Conducted research at the intersection of machine learning and bioinformatics, contributing to the MLBioIGE project focused on identifying genetic factors associated with pulmonary fibrosis in COVID-19 patients. Processed and analyzed genomic datasets using statistical and predictive modeling techniques while collaborating with multidisciplinary researchers on study design, data analysis, and publication development.

- Co-investigated MLBioIGE, integrating machine learning and bioinformatics to identify genetic factors influencing pulmonary fibrosis in COVID-19 patients; led to a peer-reviewed publication
- Processed and analyzed genomic and biomedical datasets using statistical techniques, feature extraction, and predictive modeling
- Collaborated with a multidisciplinary team on study design, data analysis, and research documentation

PUBLICATIONS / RESEARCH EXPERIENCE

- MLBioIGE: Integration and interplay of machine learning and bioinformatics to identify the genetic effect of SARS-CoV-2 on idiopathic pulmonary fibrosis patients. Biology Methods and Protocols, May 2022. https://doi.org/10.1093/biomethods/bpac013
- Enhancing the programming sequence for undergraduate computer science students: A program for improving learning outcomes. Proceedings of the 5th HCIRA International Congress, Jun 2023. Volume 7 Issue 1. DOI: 10.1109/HORA58378.2023.10156793
- Soccer Action Detection Using TimeSformer: A Transformer-Based Approach to Video Understanding. CCSC Mid-South Spring Conference 2026; Journal of Computing Sciences in Colleges, 41(10), 72-81, 2026. DOI: 10.5555/3820620.3820634
- Enhancing Disease Detection in South Asian Freshwater Fish Aquaculture Through Convolutional Neural Networks. ATU Scholars Symposium, Apr 2024. https://orc.library.atu.edu/atu_rs/2024/2024/27
- League of Learning: Deep Learning for Soccer Action Video Classification. ATU Scholars Symposium, Apr 2024. https://orc.library.atu.edu/atu_rs/2024/2024/17
- Beyond the Output: Faithful XAI for Medical Image Report Generation. IEEE BIBM 2026 Workshop on Explainable and Trustworthy AI for Clinical Decision Support Systems, Dallas, Dec 1–4, 2026. Accepted for presentation. Paper ID S55203.

PROJECTS / PORTFOLIO

Beyond the Output: Faithful XAI for Medical Image Report Generation

Researcher

Accepted for presentation at the IEEE BIBM 2026 Workshop on Explainable and Trustworthy AI for Clinical Decision Support Systems, Dallas, Dec 1–4, 2026 (Paper ID S55203; with Shucheta Bari). Investigating hallucination detection and explanation faithfulness in medical vision-language models using chest X-ray report generation as a testbed. Designing a unified evaluation framework linking visual saliency maps, model reasoning chains, and clinical correctness metrics to quantify and reduce unfaithful explanations.

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- Designing a unified evaluation framework linking visual saliency maps, model reasoning chains, and clinical correctness metrics to quantify and reduce unfaithful explanations

PyTorch, TensorFlow, Vision Transformers, Medical Imaging, XAI

Soccer Action Detection Using TimeSformer

Researcher

Achieved 91.13% Top-1 classification accuracy by fine-tuning a pretrained TimeSformer on ~3,600 annotated clips from the SoccerAct10 dataset. Developed transformer-based pipelines for automated sports analytics and fine-grained action recognition in broadcast soccer video.

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- Developing transformer-based pipelines for automated sports analytics and fine-grained action recognition in broadcast soccer video
- Modeled long-range spatiotemporal dependencies in broadcast soccer video using divided space-time attention; results accepted at CCSC Mid-South 2026

TimeSformer, PyTorch, Video Understanding, Transformer Architecture

Image Forgery Detection — ResNet50 + U-Net + Grad-CAM

Researcher

Built an explainable computer vision pipeline for detecting and localizing image manipulation, achieving AUC 0.86 and recall 0.88. Integrated Grad-CAM to produce pixel-level interpretability maps, making predictions auditable for downstream forensic use.

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- Integrated Grad-CAM to produce pixel-level interpretability maps, making predictions auditable for downstream forensic use

ResNet50, U-Net, Grad-CAM, PyTorch, Computer Vision, XAI

NewsLogic: AI-Powered Fake News Detection System

Researcher

Designed and deployed a full-stack AI system combining BERT, DistilBERT, and rule-based validation for multi-signal fake news classification. Built FastAPI backend with real-time explanation generation using LLaMA-based models, enabling transparent inference for end users.

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- Built FastAPI backend with real-time explanation generation using LLaMA-based models, enabling transparent inference for end users

BERT, DistilBERT, LLaMA, FastAPI, Python, NLP

AWARDS AND HONORS

2nd Place — Graduate Oral Presentation, ATU Research Symposium — Jan 2024

Awarded for outstanding graduate-level oral presentation at the Arkansas Tech University Research Symposium.

League of Learning: Deep Learning for Soccer Action Video Classification

1st Place — Graduate Poster Presentation, Arkansas Academy of Science — Jan 2023

Recognized for excellence in graduate-level poster presentation at the Arkansas Academy of Science conference.

League of Learning: Deep Learning for Soccer Action Video Classification